

Vol 12 Chapter 5 — Reaction Kinetics and Transition States

Keeper (author pass — deep math/physics revision)

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Chapter 5 — Reaction Kinetics and Transition States

Level 1 — one sentence

Reaction kinetics describes how fast chemical reactions proceed, with Arrhenius rate law $k = Ae^{-E_a/RT}$ giving exponential temperature dependence and Eyring transition state theory $k = (k_B T/h) \exp(-\Delta G^\ddagger/RT)$ providing the substrate-K-type-saddle interpretation: the substrate K-type configuration passes through a transient saddle-point (transition state) to reach products.

Level 2 — graduate-physicist precision

5.1 Rate laws

Empirical rate law: $r = k[A]^a[B]^b$ with reaction order $a + b$.

First-order: $r = k[A] \rightarrow$ exponential decay $[A](t) = [A]_0 e^{-kt}$. Half-life $t_{1/2} = \ln 2/k$.

Second-order: $r = k[A]^2 \rightarrow 1/[A] - 1/[A]_0 = kt$. Half-life depends on $[A]_0$.

5.2 Arrhenius equation

$$k(T) = A \exp(-E_a/RT)$$

with A pre-exponential factor (related to collision frequency, orientation), E_a activation energy.

Empirical: most reactions show Arrhenius behavior over moderate T range. Plot $\ln k$ vs $1/T \rightarrow$ slope $= -E_a/R$.

For $E_a \sim 100$ kJ/mol (typical organic chemistry): rate doubles per ~ 10 K temperature increase near room temperature.

5.3 Transition state theory (Eyring)

Eyring 1935: reaction passes through activated complex (transition state) at saddle point of potential energy surface.

$$k = \kappa \cdot \frac{k_B T}{h} \exp(-\Delta G^\ddagger/RT)$$

with κ transmission coefficient (often ~ 1), $\Delta G^\ddagger$ activation free energy.

Decomposing: $\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$. Different from Arrhenius E_a by $\Delta S^\ddagger$ contribution.

5.4 Substrate-mechanism reading

Transition state = substrate K-type saddle-point configuration. Activation energy = substrate K-type energy gap from reactant to saddle.

Catalysts work by providing alternative substrate K-type path with lower saddle. Enzymes (next chapter) lower activation energy by $\sim 10^7$ vs uncatalyzed.

5.5 K-audit anchors

- Vol 6 Ch 11: non-equilibrium thermodynamics
- Vol 8 Ch 11: chaos/dynamical systems (related transition state perspective)

Level 3 — 5th-grader accessibility

Reaction kinetics: how fast reactions go. Arrhenius: $k = Ae^{-E_a/RT}$ — exponential T dependence. Higher T $\rightarrow$ faster. Activation energy E_a = barrier. Transition state theory (Eyring): reaction passes through high-energy saddle point. Catalysts lower the saddle $\rightarrow$ speed up reaction. Enzymes are biological catalysts that speed reactions by $\sim 10^7$. In BST: transition state = substrate K-type saddle configuration; catalyst = alternative substrate K-type path.

What comes next

Chapter 6 develops spectroscopy as substrate cartography.

Where to look this up

- Steinfeld-Francisco-Hase, *Chemical Kinetics and Dynamics*
- BST: Vol 6 Ch 11